

# LCA 3 – Setting Specifications

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# Specifications

- Normally an LCA is conducted where the product and method of manufacture is known, and all the quantities of materials, etc are known
- At the moment your projects are largely unconstrained (?)
- Open ended, complex problem, with almost infinite possibilities
- Basically a design problem coupled with an LCA
- Unsolvable



# Specifications

- You want to gradually add constraints to start converting a complex problem into one that can be solved
- The first place to start is by setting some specifications for your part of the project: e.g.
- Electricity generation: max capacity (MW), type of generation, efficiency, fuel type, location, cooling, size, number of turbines, plant life time (boilers, turbines, pumps), maintenance periods
- Home heating: heat output (kW), type of heating, size, coefficient of performance, building orientation (passive heating), product life time
- Electric vehicles: type, size, weight, speed, acceleration, capacity, number of wheels, number of motors, vehicle/battery/motor life time

# Specifications

- Drinking water: Volume per day, method of distribution, method of treatment, location of plant, raw water quality, drinking water quality, plant life time
- Packaging: purpose, size, weight, thickness, material type (type of glass, plastic, paper, fabric), toughness, biodegradability, single/multiple use
- Housing: floor area, number of levels, foundation type, frame type, cladding, roofing, garage (attached or unattached), number of rooms, number of bathrooms (toilet separate from bathroom), room sizes, general plan, house life time (cladding, roofing, plumbing)
- Bridges: type, size (length, height, width), loading, construction materials, location, bridge life time, maintenance periods

# Specifications

- Waste water treatment: capacity, type of treatment, treatment steps, raw water quality (total suspended solids, total dissolved solids, total organic carbon, total nitrogen, total phosphorous, e-coli count, biological oxygen demand, chemical oxygen demand), treated water quality, location, plant life time
- Time keeping: type of watch, size, weight, materials of construction, life time

# Specifications – e.g. waste water treatment

| Parameter                               | Before treatment   | After treatment     | Tunisian standards (N.T.106.002) |
|-----------------------------------------|--------------------|---------------------|----------------------------------|
| TDS (mg/L)                              | 1,350              | 1,250               | –                                |
| Salinity (mg/L)                         | 1                  | 1                   | –                                |
| Conductivity (mS/cm)                    | $2.21 \pm 0.40^a$  | $2.14 \pm 0.15^a$   | –                                |
| Temperature (°C)                        | $35.70 \pm 4.78^a$ | $29.36 \pm 4.78^a$  | 25                               |
| pH                                      | $7.25 \pm 0.33^a$  | $7.55 \pm 0.33^a$   | $6.5 < \text{pH} < 8.5$          |
| N-NH <sub>4</sub> <sup>+</sup> (mg/L)   | $39.14 \pm 1.98^a$ | $1.65 \pm 0.78^a$   | 1                                |
| N-NO <sub>2</sub> <sup>-</sup> (mg/L)   | $4.4 \pm 1.88^a$   | $6.43 \pm 1.51^a$   | 0.5                              |
| N-NO <sub>3</sub> <sup>-</sup> (mg/L)   | $1.32 \pm 0.16^a$  | $13.4 \pm 6.26^a$   | 50                               |
| BOD <sub>5</sub> (mgO <sub>2</sub> /L)  | $80 \pm 4.58^a$    | $26 \pm 5.30^a$     | 30                               |
| COD (mgO <sub>2</sub> /L)               | $146 \pm 27.5^a$   | $45.00 \pm 28.00^a$ | 90                               |
| TSS (mg/L)                              | $40.0 \pm 2^a$     | $10.67 \pm 2.31^a$  | 30                               |
| Fecal coliforms (MPN/100 mL)            | 5,966.66           | 1,000               | 2,000                            |
| <i>Faecal streptococci</i> (MPN/100 mL) | 110,000            | $320 \pm 34.64$     | 1,000                            |

Comparison of the average wastewater quality before and after treatment to Tunisian standards  $\pm$ , standard deviations <sup>a</sup> Values are given as a mean of three replications

# How to set the specifications?

- Look up what is typical – e.g. Google (existing products, processes, infrastructure), Wikipedia, Branz, Drinking Water NZ Standards, waste water treatment discharge standards, other LCA studies
- E.g. Huntly Power Station:
  - 3x 250 MW Rankine turbines, 30% efficiency, pulverised coal fired boilers, Indonesian coal, 5% ash content, calorific value  $\sim 30$  MJ/kg, water cooled condensers, 50-60 year life time, maintenance on turbine and boiler once every 3 years
  - 1x 400 MW combined cycle (Brayton turbine plus Rankine turbine), 50% efficiency, natural gas,  $\sim 50$  MJ/kg, air cooled, maintenance once every 4 years

# How to set the specifications?

- E.g. Tesla model S: Size: 5 m by 2.2 m by 1.44 m, wheelbase 3 m, capacity 5 people, 1020 horsepower, range 412 miles, acceleration 0-60 mph in 2.1 seconds, weight 2.2 tonnes, front and rear AC induction motors
- If unknown, set a specification (until you find information)
- How detailed? Depends on the goal and scope of the LCA and how much time you have, but start off with a high level specification first and gradually add detail as you need it to progress the project: e.g.
  - Huntly – cooling water demand, typical operating capacity
  - Tesla – tires – size, weight, composition

# Example specification sheet – e.g. Huntly

| Specification                                                          | Value                                                      | Unit              |
|------------------------------------------------------------------------|------------------------------------------------------------|-------------------|
| Location                                                               | Huntly                                                     | -                 |
| Turbine type                                                           | Rankine (HP, IP, LP)                                       | -                 |
| Number of turbines                                                     | 3                                                          | -                 |
| Generation capacity per turbine                                        | 250                                                        | MW                |
| Boiler type                                                            | Suspended, tangential flow, pulverised coal, 1 per turbine | -                 |
| Fuel type                                                              | Coal - bituminous                                          | -                 |
| Fuel source                                                            | Kalimantan, Indonesia                                      | -                 |
| Coal processing equipment                                              | Grizzly, pulveriser (4 per boiler)                         | -                 |
| Boiler exhaust cleaning equipment                                      | 40 kV electrostatic precipitators (2 per boiler)           | -                 |
| Calorific value of fuel                                                | 25                                                         | MJ/kg             |
| Fuel composition (ash)                                                 | 5                                                          | %                 |
| Boiler and turbine efficiency                                          | 33                                                         | %                 |
| Boiler heat recovery                                                   | Superheater, primary, reheater, economiser, air preheater  | -                 |
| Coal demand (full load) per boiler                                     | 30.3                                                       | kg/s              |
| Cooling water demand (full load) per turbine (5 °C temperature change) | 24.3                                                       | m <sup>3</sup> /s |
| Cooling water supply                                                   | Waikato river                                              | -                 |
| Air demand (full load) per boiler                                      | 337.8                                                      | m <sup>3</sup> /s |
| Typical annual operating capacity                                      | 65                                                         | %                 |
| Building area (m2)                                                     | 44300                                                      | m <sup>2</sup>    |
| Site area (m2)                                                         | 620000                                                     | m <sup>2</sup>    |

# Activity

- Using the example specification sheet on Moodle, develop a specification sheet for your product, process, infrastructure